ECE238L Computer Logic Design

Lab 7: More State Machines and Counters

Section 003: Wednesday 2:00-4:45

Name: Tomas Salaz

One Hot Counter VHDL Source Code, Simulation Code, and Constraints

OneHotTop.vhd

C:/Users/tsalaz25/ECE238L/OneHotCounter/OneHotCounter.srcs/sources_1/new/OneHotTop.vhc

```vhdl
-- Target Devices:
-- Tool Versions:
-- Description:
--
-- Dependencies:
--
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
--
----------------------------------------------------------------------------------


library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity OneHotTop is
    Port ( Clock_System : in STD_LOGIC;
           Reset : in STD_LOGIC;
           Direction : in STD_LOGIC;
           Count : out STD_LOGIC_VECTOR (5 downto 0));
end OneHotTop;

architecture Behavioral of OneHotTop is
  component FreqDivider
    Port (Clock_System : in STD_LOGIC;
          Clock_1Hz : out STD_LOGIC);
  end component;

  component OneHotCounter
    Port (Clock_System : in STD_LOGIC;
          Direction : in STD_LOGIC;
          Reset : in STD_LOGIC;
          Count : out STD_LOGIC_VECTOR (5 downto 0));
  end component;

  signal slowClock : STD_LOGIC := '0';

begin
  fd : FreqDivider port map (Clock_System => Clock_System,
                             Clock_1Hz=> slowClock);
  hoc: OneHotCounter port map (Clock_System => slowClock,Reset=> Reset,
                               Direction => Direction,Cout => Count);
end Behavioral;
```

OneHotCounter.vhd

C:/Users/tsalaz25/ECE238L/OneHotCounter/OneHotCounter.srcs/sources_1/nev

```vhdl
entity OneHotCounter is
    Port ( Clock_System : in STD_LOGIC;
           Direction : in STD_LOGIC;
           Reset : in STD_LOGIC;
           Cout : out STD_LOGIC_VECTOR (5 downto 0));
end OneHotCounter;

architecture Behavioral of OneHotCounter is

  type state_type is (s1, s2, s4, s8, s16, s32);
  signal presState, nextState : state_type;

  begin
    process (Clock_System,Reset)
    begin
      if Reset = '1' then
        presState <= s1;
      elsif rising_edge (Clock_System) then
        presState <= nextState;
      end if;
    end process;

    process (presState, Direction)
    begin
      case presState is
        when s1 =>
          Cout <= "000001";
          if Direction = '1' then nextState <= s2;
          else nextState <= s32; end if;
        when s2 =>
          Cout <= "000010";
          if Direction = '1' then nextState <= s4;
          else nextState <= s1; end if;
        when s4 =>
          Cout <= "000100";
          if Direction = '1' then nextState <= s8;
          else nextState <= s2; end if;
        when s8 =>
          Cout <= "001000";
          if Direction = '1' then nextState <= s16;
          else nextState <= s4; end if;
        when s16 =>
          Cout <= "010000";
          if Direction = '1' then nextState <= s32;
          else nextState <= s8; end if;
        when s32 =>
          Cout <= "100000";
          if Direction = '1' then nextState <= s1;
          else nextState <= s16; end if;
       end case;
     end process;
end Behavioral;
```

FreqDivider.vhd

C:/Users/tsalaz25/ECE238L/FreqDivider/FreqDivider.srcs/sources_1/new/FreqDivider.vhd

```vhdl
-- Create Date: 03/11/2025 01:26:10 PM
-- Design Name:
-- Module Name: FreqDivider - Behavioral
-- Project Name:
-- Target Devices:
-- Tool Versions:
-- Description:
--
-- Dependencies:
--
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
--
----------------------------------------------------------------------------------


library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity FreqDivider is
    Port ( Clock_System : in STD_LOGIC;
           Clock_1Hz : out STD_LOGIC);
end FreqDivider;

architecture Behavioral of FreqDivider is
signal count : INTEGER := 1 ;
signal sClock_1Hz : STD_LOGIC := '0';

begin
process (Clock_System)
begin
  if Clock_System' event and Clock_System = '1' then
    count <= count + 1;
    if count = 50000000 then                  --LINE FOR CHANGING
      Clock_1Hz <= not sClock_1Hz;
      sClock_1Hz <= not sClock_1Hz;
      count <= 1;
    end if;
  end if;
end process;
end Behavioral;
```

OneHotCounter_tb.vhd

C:/Users/tsalaz25/ECE238L/OneHotCounter/OneHotCounter.srcs/sim_1/new/OneHotCounter_tb.vl

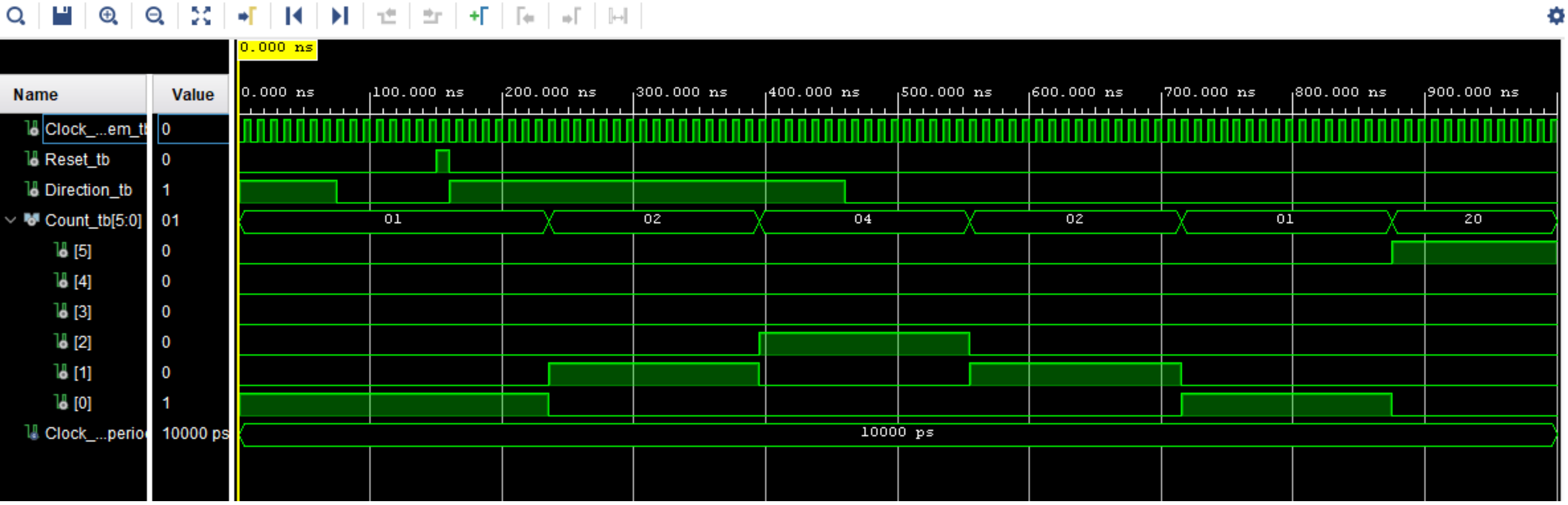
```vhdl
-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity OneHotCounter_tb is
--  Port ( );
end OneHotCounter_tb;

architecture Bench of OneHotCounter_tb is
  component OneHotTop
    Port ( Clock_System : in STD_LOGIC;
           Reset : in STD_LOGIC;
           Direction : in STD_LOGIC;
           Count : out STD_LOGIC_VECTOR (5 downto 0));
  end component;

  signal Clock_System_tb : STD_LOGIC;
  signal Reset_tb : STD_LOGIC;
  signal Direction_tb : STD_LOGIC;
  signal Count_tb : STD_LOGIC_VECTOR(5 downto 0);
  constant Clock_1000MHz_period : time := 10ns;

  begin
   uut: OneHotTop Port map (Clock_System => Clock_System_tb,
                            Reset => Reset_tb,
                            Direction => Direction_tb,
                            Count => Count_tb);
  clk_process : process
    begin
      Clock_System_tb <= '0'; wait for Clock_1000MHz_period/2;
      Clock_System_tb <= '1'; wait for Clock_1000MHz_period/2;
    end process;

   stimulus : process
    begin
      Reset_tb <= '0';
      Direction_tb <= '1'; wait for 75 ns;
      Direction_tb <= '0'; wait for 75 ns;
      Reset_tb <= '1'; wait for 10 ns;
      Reset_tb <= '0';
      Direction_tb <= '1'; wait for 300 ns;
      Direction_tb <= '0'; wait for 300 ns;


      wait;
    end process;
end Bench;
```

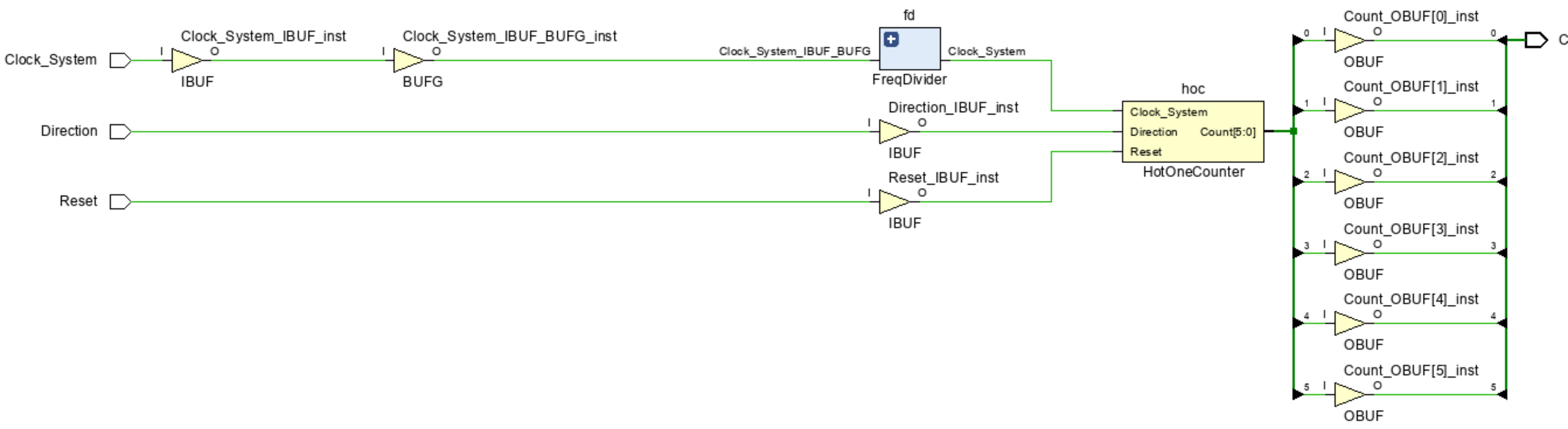
OneHot_Constr.xdc

C:/Users/tsalaz25/ECE238L/OneHotCounter/OneHotCounter.srcs/constrs_1/new/OneHot_Constr.›

```tcl
set_property IOSTANDARD LVCMOS33 [get_ports {Count[5]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Count[4]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Count[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Count[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Count[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Count[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports Clock_System]
set_property IOSTANDARD LVCMOS33 [get_ports Direction]
set_property IOSTANDARD LVCMOS33 [get_ports Reset]
set_property PACKAGE_PIN E3 [get_ports Clock_System]
set_property PACKAGE_PIN J15 [get_ports Reset]
set_property PACKAGE_PIN L16 [get_ports Direction]
set_property PACKAGE_PIN H17 [get_ports {Count[0]}]
set_property PACKAGE_PIN K15 [get_ports {Count[1]}]
set_property PACKAGE_PIN J13 [get_ports {Count[2]}]
set_property PACKAGE_PIN N14 [get_ports {Count[3]}]
set_property PACKAGE_PIN R18 [get_ports {Count[4]}]
set_property PACKAGE_PIN V17 [get_ports {Count[5]}]
```

One Hot Counter Waveform and Schematic





Ten State Counter VHDL Source Code, Simulation Code and Constraints

FreqDivider.vhd

C:/Users/tsalaz25/ECE238L/FreqDivider/FreqDivider.srcs/sources_1/ne

```vhdl
-- Create Date: 03/11/2025 01:26:10 PM
-- Design Name:
-- Module Name: FreqDivider - Behavioral
-- Project Name:
-- Target Devices:
-- Tool Versions:
-- Description:
--
-- Dependencies:
--
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
--
----------------------------------------------------------------------------


library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if us
-- arithmetic functions with Signed or Unsigned valu
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if in
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity FreqDivider is
    Port ( Clock_System : in STD_LOGIC;
           Clock_1Hz : out STD_LOGIC);
end FreqDivider;

architecture Behavioral of FreqDivider is
signal count : INTEGER := 1 ;
signal sClock_1Hz : STD_LOGIC := '0';

begin
process (Clock_System)
begin
  if Clock_System' event and Clock_System = '1' then
    count <= count + 1;
    if count = 50000000 then                        --LIN
      Clock_1Hz <= not sClock_1Hz;
      sClock_1Hz <= not sClock_1Hz;
      count <= 1;
    end if;
  end if;
end process;
end Behavioral;
```

TenStateCounter.vhd

C:/Users/tsalaz25/ECE238L/TenStateCounter/TenStateCounter.srcs/sources_1/ne

```vhdl
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiat
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity TenStateCounter is
    Port ( Clock_System : in STD_LOGIC;
           Direction : in STD_LOGIC;
           Reset : in STD_LOGIC;
           Count : out STD_LOGIC_VECTOR (3 downto 0));
end TenStateCounter;

architecture Behavioral of TenStateCounter is

  type state_type is (s0, s1, s2, s3, s4, s5, s6, s7, s8, s9
  signal presentState, nextState : state_type;

  begin
  syncProc : process (Clock_System, Reset)
  begin
    if Reset = '1' then
      presentState <= s0;
    elsif rising_edge (Clock_System) then
      presentState <= nextState;
    end if;
  end process;

  comboProc : process (presentState, Direction)
  begin
    case presentState is
      when s0 => Count <= "0000";
        if Direction = '1' then nextState <= s1;
        else nextState <= s9;
        end if;
      when s1 => Count <= "0001"; nextState <= s2;
      when s2 => Count <= "0010"; nextState <= s3;
      when s3 => Count <= "0011"; nextState <= s4;
      when s4 => Count <= "0100"; nextState <= s0;
      when s5 => Count <= "0101"; nextState <= s9;
      when s6 => Count <= "0110"; nextState <= s5;
      when s7 => Count <= "0111"; nextState <= s6;
      when s8 => Count <= "1000"; nextState <= s7;
      when s9 => Count <= "1001";
        if Direction = '0' then nextState <= s8;
        else nextState <= s0;
        end if;
    end case;
  end process;
end Behavioral;
```

DisplayDigits.vhd

C:/Users/tsalaz25/ECE238L/DisplayDigits/DisplayDigits.srcs/sources_1/new/Disp

```vhdl
-- Dependencies:
--
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
--
----------------------------------------------------------------------------


library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instanti
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity DisplayDigits is
    Port ( Count : in STD_LOGIC_VECTOR (3 downto 0);
           Cathode_7SD : out STD_LOGIC_VECTOR (7 downto 0)
           Anode_7SD : out STD_LOGIC_VECTOR (7 downto 0));
end DisplayDigits;

architecture Behavioral of DisplayDigits is
begin
  process (Count)
  begin
    if Count = "0000" then Cathode_7SD <= "00000011" ; --0
    elsif Count = "0010" then Cathode_7SD <= "10011111" ;
    elsif Count = "0010" then Cathode_7SD <= "00100101" ;
    elsif Count = "0011" then Cathode_7SD <= "00001101" ;
    elsif Count = "0100" then Cathode_7SD <= "10011001" ;
    elsif Count = "0101" then Cathode_7SD <= "01001001" ;
    elsif Count = "0110" then Cathode_7SD <= "11000001" ;
    elsif Count = "0111" then Cathode_7SD <= "00011111" ;
    elsif Count = "1000" then Cathode_7SD <= "00000001" ;
    elsif Count = "1001" then Cathode_7SD <= "00001001" ;

    elsif Count = "1010" then Cathode_7SD <= "00010001" ;
    elsif Count = "1011" then Cathode_7SD <= "11000001" ;
    elsif Count = "1100" then Cathode_7SD <= "01100011" ;
    elsif Count = "1101" then Cathode_7SD <= "10000101" ;
    elsif Count = "1110" then Cathode_7SD <= "01100001" ;
    elsif Count = "1111" then Cathode_7SD <= "01110001" ;
  end if;
    Anode_7SD <= "11111110";
  end process;
end Behavioral;
```

TenStateTop.vhd

C:/Users/tsalaz25/ECE238L/TenStateCounter/TenStateCounter.srcs/sources_1/new/TenSta

```vhdl
--use UNISIM.VComponents.all;

entity TenStateTop is
    Port (
        Clock_System : in STD_LOGIC;
        Direction    : in STD_LOGIC;
        Reset        : in STD_LOGIC;
        Cathode7SD   : out STD_LOGIC_VECTOR(7 downto 0);
        Anode7SD     : out STD_LOGIC_VECTOR(7 downto 0)
    );
end TenStateTop;

architecture Behavioral of TenStateTop is
    component FreqDivider
        Port (
            Clock_System : in STD_LOGIC;
            Clock_1Hz    : out STD_LOGIC);
    end component;
    component DisplayDigits
        Port (
            Count        : in STD_LOGIC_VECTOR(3 downto 0);
            Cathode_7SD  : out STD_LOGIC_VECTOR(7 downto 0);
            Anode_7SD    : out STD_LOGIC_VECTOR(7 downto 0));
    end component;
    component TenStateCounter
        Port (
            Clock_System : in STD_LOGIC;
            Direction     : in STD_LOGIC;
            Reset         : in STD_LOGIC;
            Count         : out STD_LOGIC_VECTOR(3 downto 0));
    end component;

    signal slowClock : STD_LOGIC := '0';
    signal Count     : STD_LOGIC_VECTOR(3 downto 0) := "0000";

begin
    FD : FreqDivider
        port map (
            Clock_System => Clock_System,
            Clock_1Hz    => slowClock);

    TSC : TenStateCounter port map (
            Clock_System => slowClock,
            Reset        => Reset,
            Direction    => Direction,
            Count        => Count);

    DD : DisplayDigits port map (
            Count        => Count,
            Cathode_7SD  => Cathode7SD,
            Anode_7SD    => Anode7SD );
end Behavioral;
```

Ten State Counter Code Continued

TenState_tb.vhd

C:/Users/tsalaz25/ECE238L/TenStateCounter/TenStateCounter.srcs/sim_1/new/TenState_tb.vhd

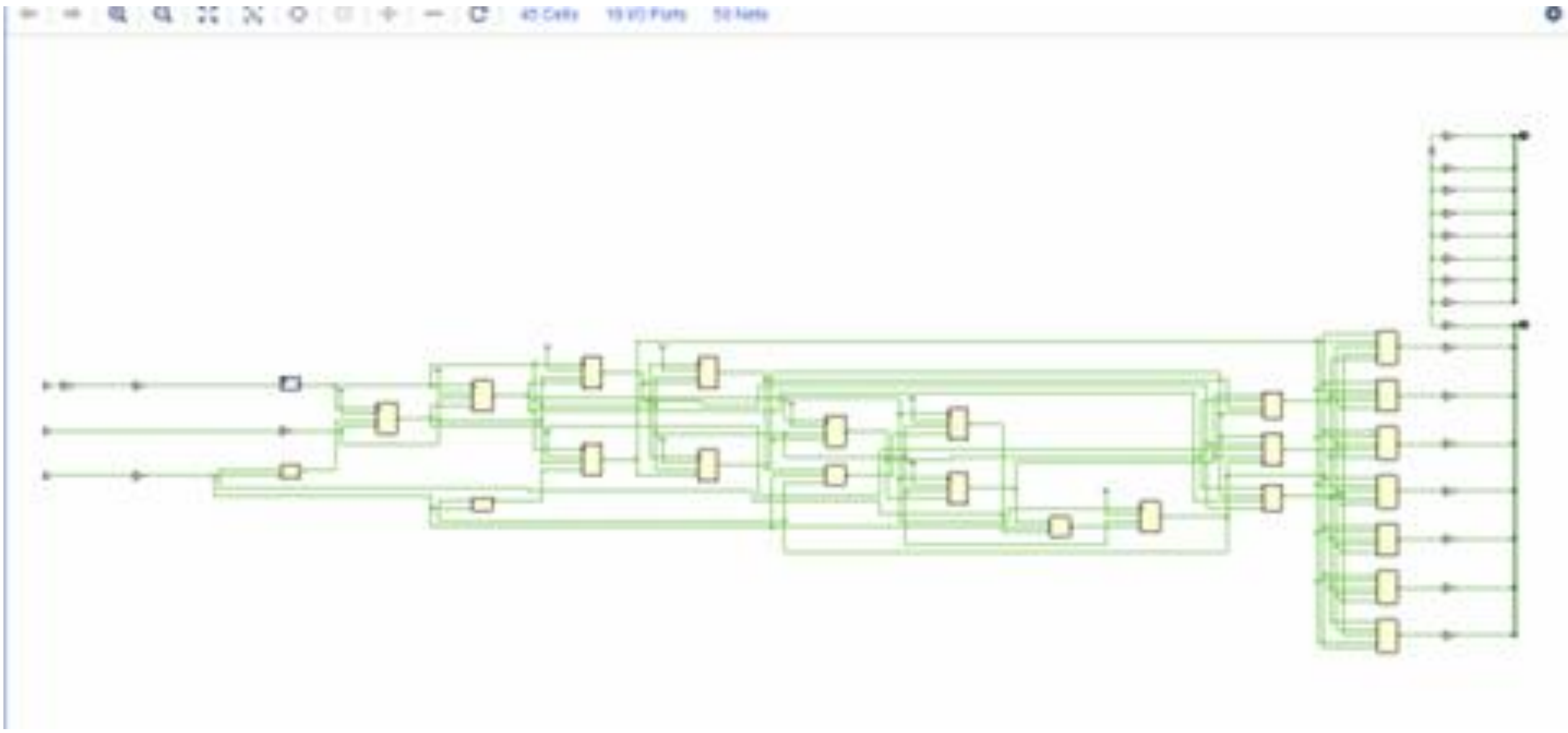
```vhdl

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity TenState_tb is
--  Port ( );
end TenState_tb;

architecture Behavioral of TenState_tb is
  component TenStateTop is
    Port (
        Clock_System : in STD_LOGIC;
        Direction    : in STD_LOGIC;
        Reset        : in STD_LOGIC;
        Cathode7SD   : out STD_LOGIC_VECTOR(7 downto 0);
        Anode7SD     : out STD_LOGIC_VECTOR(7 downto 0));
  end component;

  signal Clock_System_tb , Reset_tb, Direction_tb : STD_LOGIC;
  signal Cathode7SD_tb, Anode_7SD_tb : STD_LOGIC_VECTOR (7 downto 0);
  constant Clock_1000MHz_period : time := 15ns;

  begin
  uut : TenStateTop port map (
         Clock_System => Clock_System_tb,
         Direction    => Direction_tb   ,
         Reset        => Reset_tb       ,
         Cathode7SD   => Cathode7SD_tb  ,
         Anode7SD     => Anode_7SD_tb   );

  clk_process : process
  begin
    Clock_System_tb <= '0' ; wait for Clock_1000MHz_period/4;
    Clock_System_tb <= '1' ; wait for Clock_1000MHz_period/4;
  end process;

  direction : process
  begin
    Direction_tb <= '0'; wait for 500 ns;
    Direction_tb <= '1'; wait for 500 ns;
  end process;

  stimulus : process
  begin
    Reset_tb <= '0'; wait for 300ns;
    Reset_tb <= '0'; wait for 40ns;
    Reset_tb <= '1'; wait for 600ns;
  end process;

end Behavioral;

```

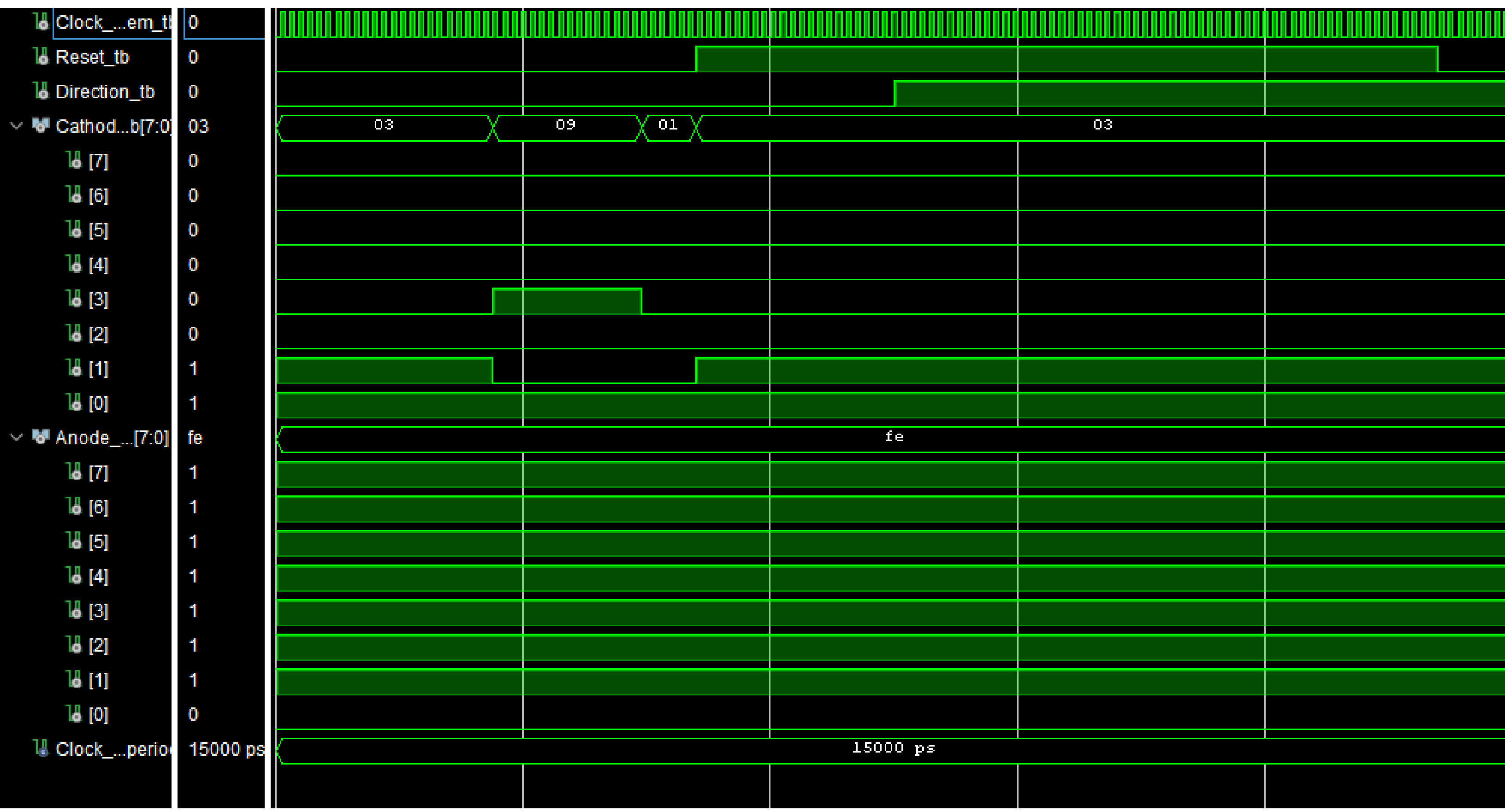
TenState_const.xdc

C:/Users/tsalaz25/ECE238L/TenStateCounter/TenStateCounter.srcs/constrs_1/ne

```
set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[7]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[6]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[5]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[4]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[7]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[6]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[5]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[4]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports Clock_System]
set_property IOSTANDARD LVCMOS33 [get_ports Direction]
set_property IOSTANDARD LVCMOS33 [get_ports Reset]
set_property PACKAGE_PIN E3 [get_ports Clock_System]
set_property PACKAGE_PIN L16 [get_ports Direction]
set_property PACKAGE_PIN J15 [get_ports Reset]
set_property PACKAGE_PIN J17 [get_ports {Anode7SD[0]}]
set_property PACKAGE_PIN J18 [get_ports {Anode7SD[1]}]
set_property PACKAGE_PIN T9 [get_ports {Anode7SD[2]}]
set_property PACKAGE_PIN J14 [get_ports {Anode7SD[3]}]
set_property PACKAGE_PIN P14 [get_ports {Anode7SD[4]}]
set_property PACKAGE_PIN T14 [get_ports {Anode7SD[5]}]
set_property PACKAGE_PIN K2 [get_ports {Anode7SD[6]}]
set_property PACKAGE_PIN U13 [get_ports {Anode7SD[7]}]
set_property PACKAGE_PIN H15 [get_ports {Cathode7SD[0]}]
set_property PACKAGE_PIN L18 [get_ports {Cathode7SD[1]}]
set_property PACKAGE_PIN T11 [get_ports {Cathode7SD[2]}]
set_property PACKAGE_PIN P15 [get_ports {Cathode7SD[3]}]
set_property PACKAGE_PIN K13 [get_ports {Cathode7SD[4]}]
set_property PACKAGE_PIN K16 [get_ports {Cathode7SD[5]}]
set_property PACKAGE_PIN R10 [get_ports {Cathode7SD[6]}]
set_property PACKAGE_PIN T10 [get_ports {Cathode7SD[7]}]
```

Ten State Counter Waveform and Schematic





Summary:
Both parts of this lab build on what we have learned since lab 5. The goal of this lab was to make more state machine counters and display the counters in more complex ways, we also added a Direction to the counting which made this lab a bit more challenging. The One Hot Counter behaved in the same way as the Grey Code Counter except that we had to implement the bit more code for the Direction functionality. The process was similar, as I implemented a Top Module with a Frequency Divider Component and a One Hot Counter Component to have the Display, the Divider and the Courting all separate.
Part B was similar, as we had to implement the same kind of counter, the differences being that this Ten State Counter was displayed on the 7-segment Display and, depending on direction, counted from 0-4 or down from 9-5. I implemented the same structure in this lab, having a Top Module for the Display, A Frequency Divider and a Counter, with the same Strucure. The only small difference is that the Tip module had "Display Digits" (lab 5c) to help with the 7 Segment Display.

Problems:
This lab took me longer than it should have, I kept making syntax mistakes and mixing up the name of the components. Other than there, there weren't any technical problems as the lab was relatively the same as the last lab.

Hints:
Attention ot Detail matters.

Recommendations:
None, It was a good lab.